

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Case Study

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I **Yarra Pawan (192424294)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y. Pawan**

Case Study Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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Total Marks Awarded:

Signature of the Course Faculty

Questions:

SET 1

1. Retail Sales Data Analysis

A retail company has collected sales data containing customer details, product categories, quantities, prices, purchase dates, and cities from different branches. Using Python and Pandas, import the dataset, explore its structure and table functions, identify missing or unusual values, create suitable groupings based on product category and city, and analyze total sales and quantity sold. Present the important findings using appropriate charts and provide a brief interpretation of the results.

2. Customer Behavior Analysis

An online shopping company wants to understand customer purchasing behavior using a dataset containing customer ID, age, gender, location, product category, order value, and purchase date. Import and explore the data, separate relevant customer groups, analyze purchasing patterns, identify relationships between customer characteristics and order value, and identify any significant outliers. Present the findings using suitable visualizations and explain how the analysis can help the company improve its marketing strategy.

3. Student Academic Performance

A college has collected student information including student ID, department, attendance percentage, assignment marks, internal marks, examination marks, and semester results. Explore the dataset using Pandas, create groups based on departments, identify correlations between attendance and academic performance, detect unusual performance values, and analyze the factors associated with higher marks. Present the results using charts and provide suitable recommendations to improve student performance.

4. Healthcare Patient Analysis

A hospital maintains patient records containing age, gender, diagnosis, treatment type, admission date, length of stay, and medical expenses. Import and explore the dataset, identify outliers in patient age and treatment expenses, group patients according to diagnosis and treatment type, and analyze relationships between length of stay and medical expenses. Create appropriate visualizations, including time-related charts where applicable, and prepare a concise presentation of the major findings.

SET 2

1. Employee Performance Analysis

An organization has collected employee data containing employee ID, department, designation, salary, experience, performance score, and joining date. Analyze the dataset by exploring its table structure, grouping employees by department and designation, identifying correlations between experience, salary, and performance, and detecting unusual salary or performance values. Present the results through appropriate charts and explain how the findings can support employee management and decision-making.

2. Banking Transaction Analysis

A bank has provided transaction data containing customer ID, account type, transaction type, transaction amount, transaction date, and branch location. Import and explore the data, identify unusually large transactions, group transactions according to account type and branch, analyze relationships between transaction frequency and transaction amount, and examine transaction trends over time. Present the findings using suitable charts and provide insights that could help the bank understand customer transaction behavior.

3. E-Commerce Product Analysis

An e-commerce company wants to identify its best-performing products using product, sales, customer, and review datasets collected from different sources. Integrate the datasets using appropriate Pandas operations, explore the resulting table, group products by category, analyze relationships between product ratings, prices, and sales, and identify products with unusually high or low sales. Create suitable visualizations and present recommendations for improving product sales.

4. Weather Data Analysis

A meteorological organization has collected daily weather information containing city, date, temperature, humidity, rainfall, and wind speed. Import and explore the dataset, analyze temperature and rainfall patterns over time, group observations by city and month, identify unusual weather observations, and investigate correlations between temperature, humidity, rainfall, and wind speed. Present the analysis using time-related charts and other suitable visualizations, and summarize the major weather patterns discovered.

SET 3

1. Transportation and Traffic Analysis

A city transportation department maintains data containing vehicle type, route, date, travel time, distance, traffic volume, and location. Explore the dataset, create groups based on vehicle type and route, identify outliers in travel time and traffic volume, and analyze relationships between distance, traffic volume, and travel time. Use time-related visualizations to identify traffic patterns and, where geographical information is available, represent important findings using maps. Prepare a report explaining how the results can support traffic management.

2. Food Delivery Analysis

A food delivery platform has collected order information including customer location, restaurant, food category, order value, delivery time, order date, and customer rating. Analyze the dataset by exploring its structure, grouping orders by restaurant and food category, identifying unusual delivery times, and determining whether order value is related to customer ratings or delivery time. Analyze changes in orders over time and present the important findings using charts and other suitable visualizations.

3. Movie and Entertainment Analysis

A streaming platform has collected information about movies and shows, including title, genre, release year, duration, language, rating, number of viewers, and country. Explore and analyze the dataset, create groups based on genre and language, identify relationships between ratings, duration, and viewership, and detect unusual values. Present trends across release years using time-related charts and use appropriate visualizations to identify the most popular genres and content categories.

4. Agriculture and Crop Analysis

An agricultural department has collected data containing crop type, district, cultivated area, rainfall, fertilizer usage, and crop yield. Import and explore the data, group the records by crop and district, identify outliers in yield and fertilizer usage, and analyze correlations between rainfall, cultivated area, fertilizer usage, and crop yield. Use charts and maps, where geographical information is available, to communicate the major findings and suggest suitable strategies for improving agricultural productivity.

SET 4

1. Social Media Engagement Analysis

A company wants to analyze the performance of its social media posts using data containing platform, post type, posting date, likes, shares, comments, views, and engagement rate. Explore the dataset, group posts according to platform and content type, identify relationships between views and engagement, detect unusual engagement values, and analyze changes in engagement over time. Create suitable charts and interactive visualizations and prepare a presentation explaining which types of content generate the highest engagement.

2. Energy Consumption Analysis

A power management organization has collected electricity consumption data containing household ID, location, date, appliance type, energy consumption, and peak usage. Explore and clean the dataset, group consumption by location and appliance type, identify unusually high energy consumption, analyze relationships between appliance usage and total consumption, and examine consumption patterns over time. Present the results using appropriate charts and, if location information is available, maps, and provide recommendations for reducing energy consumption.

3. Tourism Data Analysis

A tourism department has collected data containing tourist location, destination, travel date, accommodation type, expenditure, duration of stay, and tourist rating. Analyze the dataset by exploring its structure, grouping tourists by destination and accommodation type, identifying outliers in expenditure and duration of stay, and examining relationships between duration, expenditure, and ratings. Analyze tourism trends across different months or years and present the findings using charts, maps, and other suitable presentation techniques.

4. Integrated Business Data Analysis and Presentation

A multinational company has collected separate datasets containing customer information, sales transactions, product details, and regional information. Integrate the datasets using appropriate Pandas functions, explore the combined dataset, create meaningful groupings, identify correlations and outliers, and perform detailed analysis to determine the company's most profitable products, customer segments, and regions. Present the results using a combination of charts, time-related visualizations, maps, and interactive elements, and publish the final analysis using an appropriate presentation or open-source data visualization platform.

Question 1: Employee Performance Analysis

Understanding of the Case

An organization maintains a record of its employees containing EmployeeID, Department, Designation, Salary, Experience (in years), Performance Score and Joining Date. The goal is to explore this dataset, understand how employees are distributed across departments and designations, and determine whether experience and salary influence performance. The dataset used for this case (employee_performance.csv) contains 120 employee records across six departments (Sales, IT, HR, Finance, Marketing and Operations).

Application of Concepts

The dataset was imported using `pandas.read_csv()` and explored using `head()`, `info()` and `describe()` to understand its structure and data types. `groupby()` was used to aggregate the data by Department and by Designation to compute average salary, performance and experience. `corr()` was used to measure the linear relationship between Salary, Experience and Performance Score. The Interquartile Range (IQR) method was applied to Salary and Performance Score to flag unusual values.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv("employee_performance.csv")

dept_group = df.groupby("Department").agg( Avg_Salary=("Salary", "mean"),
    Avg_Performance=("Performance_Score", "mean"),).round(1)

corr_matrix = df[["Salary", "Experience_Years", "Performance_Score"]].corr()

fig, ax1 = plt.subplots(figsize=(9, 5))

x = range(len(dept_group))

ax1.bar(x, dept_group["Avg_Salary"], color="#4C72B0", width=0.4, label="Avg Salary")

ax1.set_ylabel("Average Salary (Rs.)")

ax1.set_xticks(x)

ax1.set_xticklabels(dept_group.index, rotation=20)

ax2 = ax1.twinx()

ax2.plot(x, dept_group["Avg_Performance"], color="#DD8452", marker="o", label="Avg Performance")

ax2.set_ylabel("Average Performance Score")

fig.legend(loc="upper right", bbox_to_anchor=(0.9, 0.9))

plt.title("Average Salary and Performance by Department")

plt.tight_layout()

plt.show()
```

```

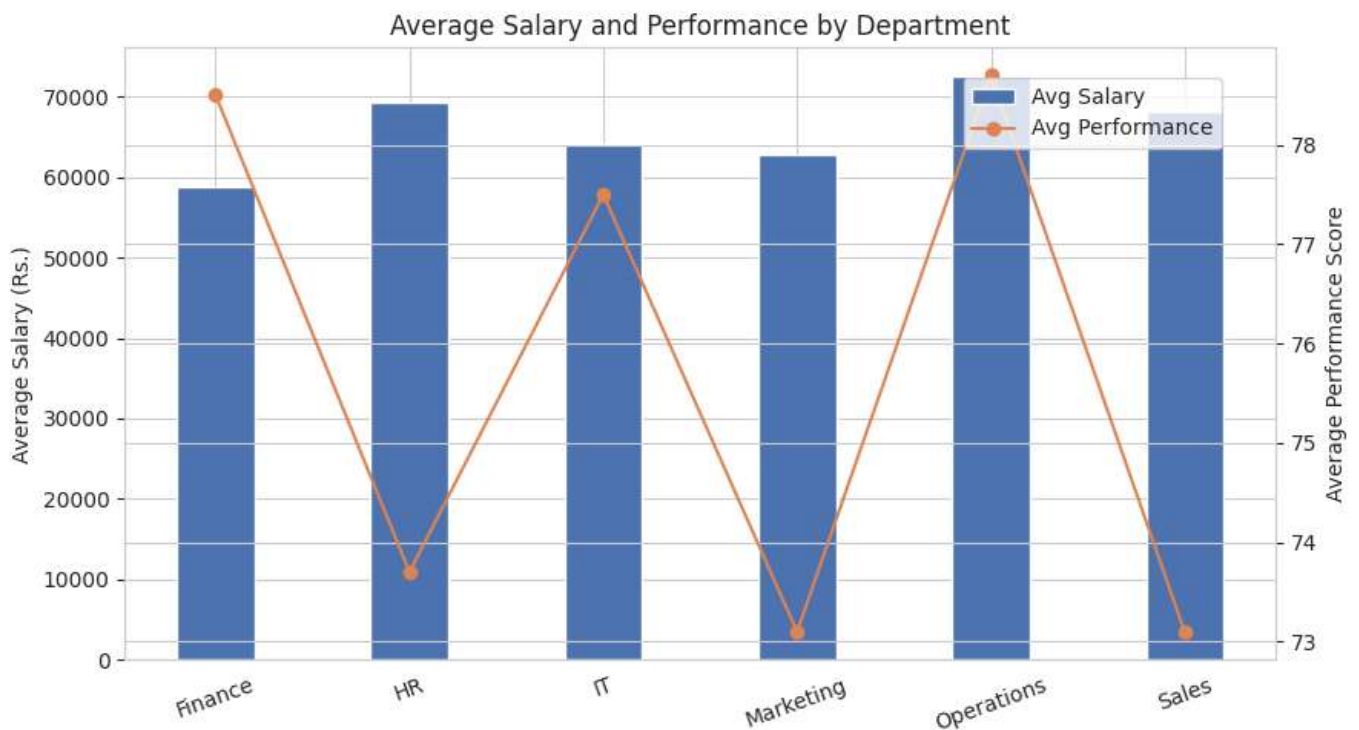
plt.figure(figsize=(8, 6))
sns.scatterplot( data=df, x="Experience_Years", y="Salary", hue="Performance_Score",
    palette="viridis", size="Performance_Score", sizes=(20, 120))
plt.title("Experience vs Salary (colored by Performance Score)")
plt.xlabel("Experience (Years)")
plt.ylabel("Salary (Rs.)")
plt.tight_layout()
plt.show()

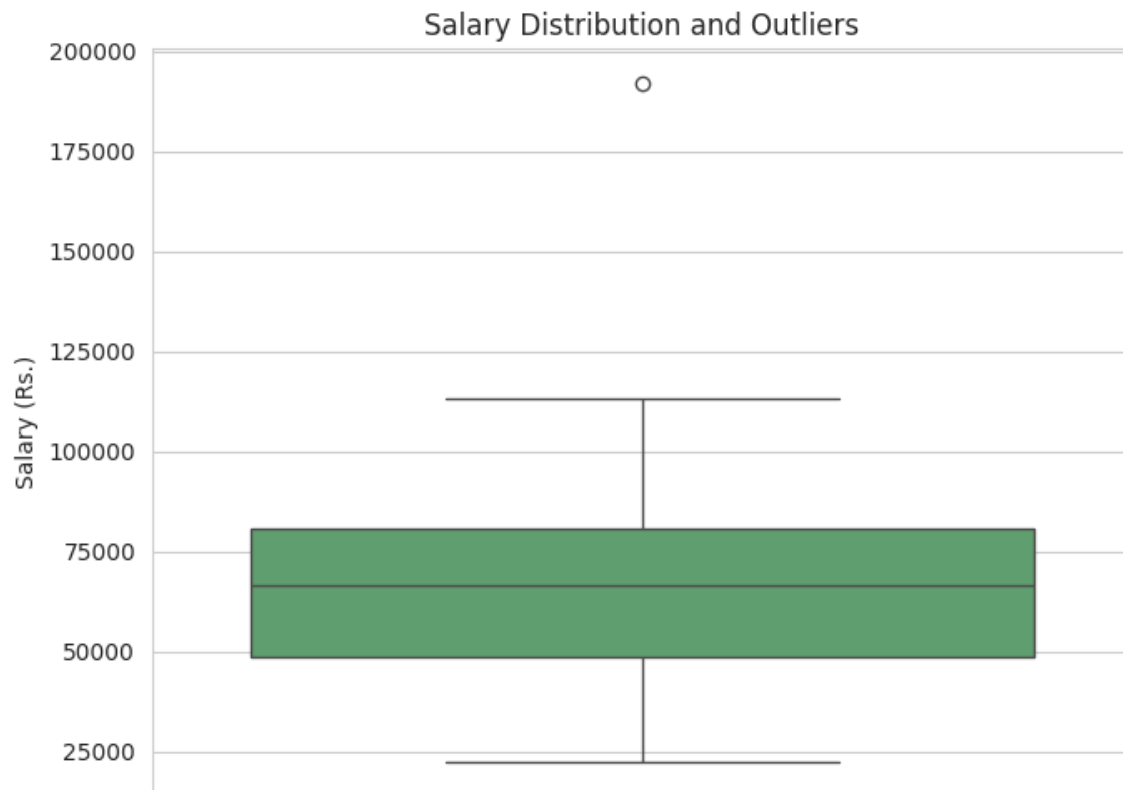
plt.figure(figsize=(7, 5))
sns.boxplot(y=df["Salary"], color="#55A868")
plt.title("Salary Distribution and Outliers")
plt.ylabel("Salary (Rs.)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(5.5, 4.5))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", vmin=-1, vmax=1, fmt=".2f")
plt.title("Correlation: Salary, Experience, Performance")
plt.tight_layout()
plt.show()

```

Output:







Analysis

The dataset had no missing values. Average salary ranged from about ₹58,800 in Finance to ₹72,500 in Operations, while average performance scores stayed close together across departments (73–79). Grouping by designation showed that Managers and Team Leads earned the highest average salaries (₹85,265 and ₹68,112 respectively) and also had the highest average performance (77.8 and 79.3). The correlation matrix showed a moderate positive relationship between Experience and Salary ($r = 0.68$), a weak positive relationship between Experience and Performance ($r = 0.31$), and a weak relationship between Salary and Performance ($r = 0.26$) — meaning salary is driven mainly by experience and designation, not purely by performance. Using the IQR method, one employee in Operations was flagged with an unusually high salary (₹1,92,080), and two employees (one in Marketing, one in HR) had unusually low performance scores (15.0), which need HR review.

Solution Design

Management can use these findings to (1) review the two employees with unusually low performance scores through targeted coaching or a performance improvement plan, (2) audit the single outlier salary record for data-entry or policy errors, (3) introduce performance-linked increments since salary currently tracks experience more than performance, and (4) prioritize retention efforts for Operations and HR, which show both good performance and competitive salaries.

Clarity

The analysis is organized department-wise and designation-wise with clear grouped summaries, a correlation matrix, and outlier detection, so the results can be read and acted upon directly by HR without further processing.

Question 2: Banking Transaction Analysis

Understanding of the Case

A bank has transaction-level data with CustomerID, Account_Type, Transaction_Type, Transaction_Amount, Transaction_Date and Branch_Location. The task is to identify unusually large transactions, understand transaction behavior across account types and branches, and study how transaction frequency relates to transaction amount over time. The dataset used (banking_transactions.csv) contains 400 transactions from January to December 2024 across five branches.

Application of Concepts

The Transaction_Date column was converted to a datetime type using pd.to_datetime(). groupby() was applied on Account_Type and Branch_Location to compute total and average transaction amounts. The IQR method was used on Transaction_Amount to detect unusually large transactions. A per-customer aggregation (count and mean amount) was used to test the correlation between transaction frequency and average amount. The date column was converted to a monthly period using dt.to_period('M') to study the transaction trend over time.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
df = pd.read_csv("banking_transactions.csv")
df["Transaction_Date"] = pd.to_datetime(df["Transaction_Date"])
acc_group = df.groupby("Account_Type")["Transaction_Amount"].sum()
branch_group = df.groupby("Branch_Location")["Transaction_Amount"].sum()
df["Month"] = df["Transaction_Date"].dt.to_period("M")
monthly_trend = df.groupby("Month")["Transaction_Amount"].sum()
plt.figure(figsize=(7, 5))
sns.barplot(x=acc_group.index, y=acc_group.values, hue=acc_group.index, palette="Blues_d",
legend=False)
plt.title("Total Transaction Amount by Account Type")
plt.ylabel("Total Amount (Rs.)")
plt.xlabel("Account Type")
plt.tight_layout()
plt.show()
plt.figure(figsize=(7, 5))
sns.barplot(x=branch_group.index, y=branch_group.values, hue=branch_group.index,
palette="Greens_d", legend=False)
```

```

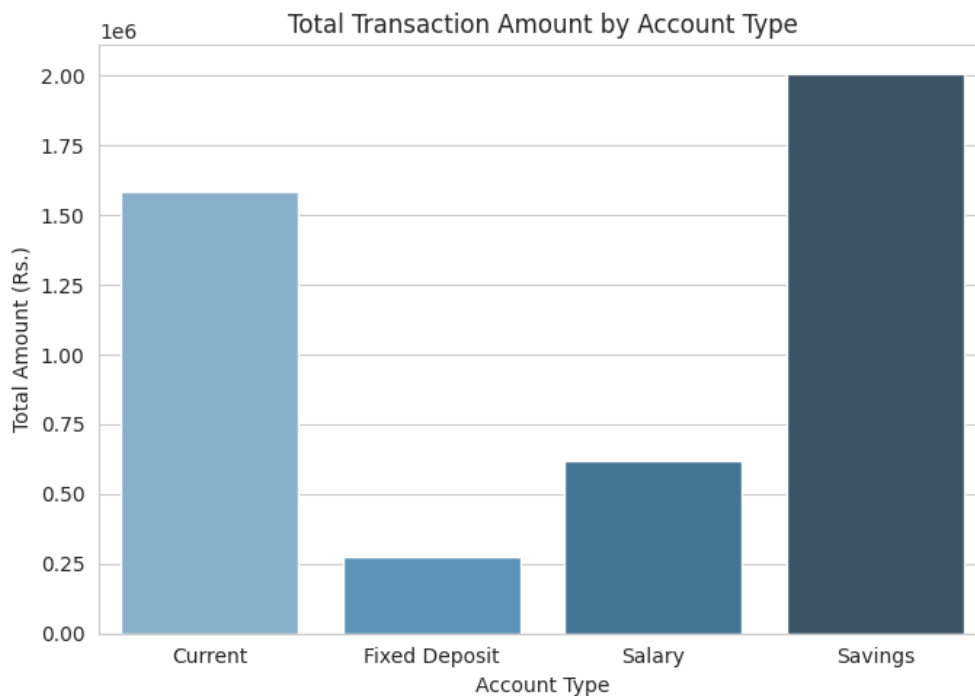
plt.title("Total Transaction Amount by Branch")
plt.ylabel("Total Amount (Rs.)")
plt.xlabel("Branch")
plt.tight_layout()
plt.show()

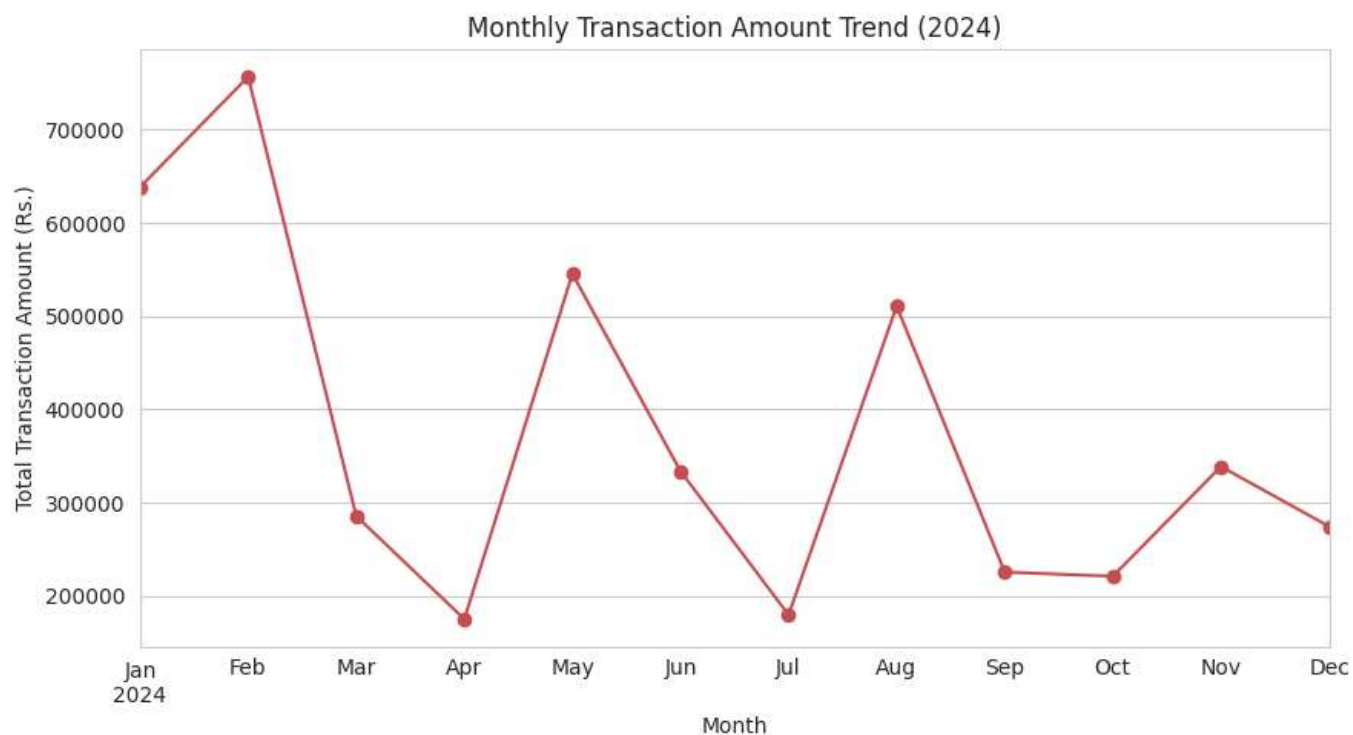
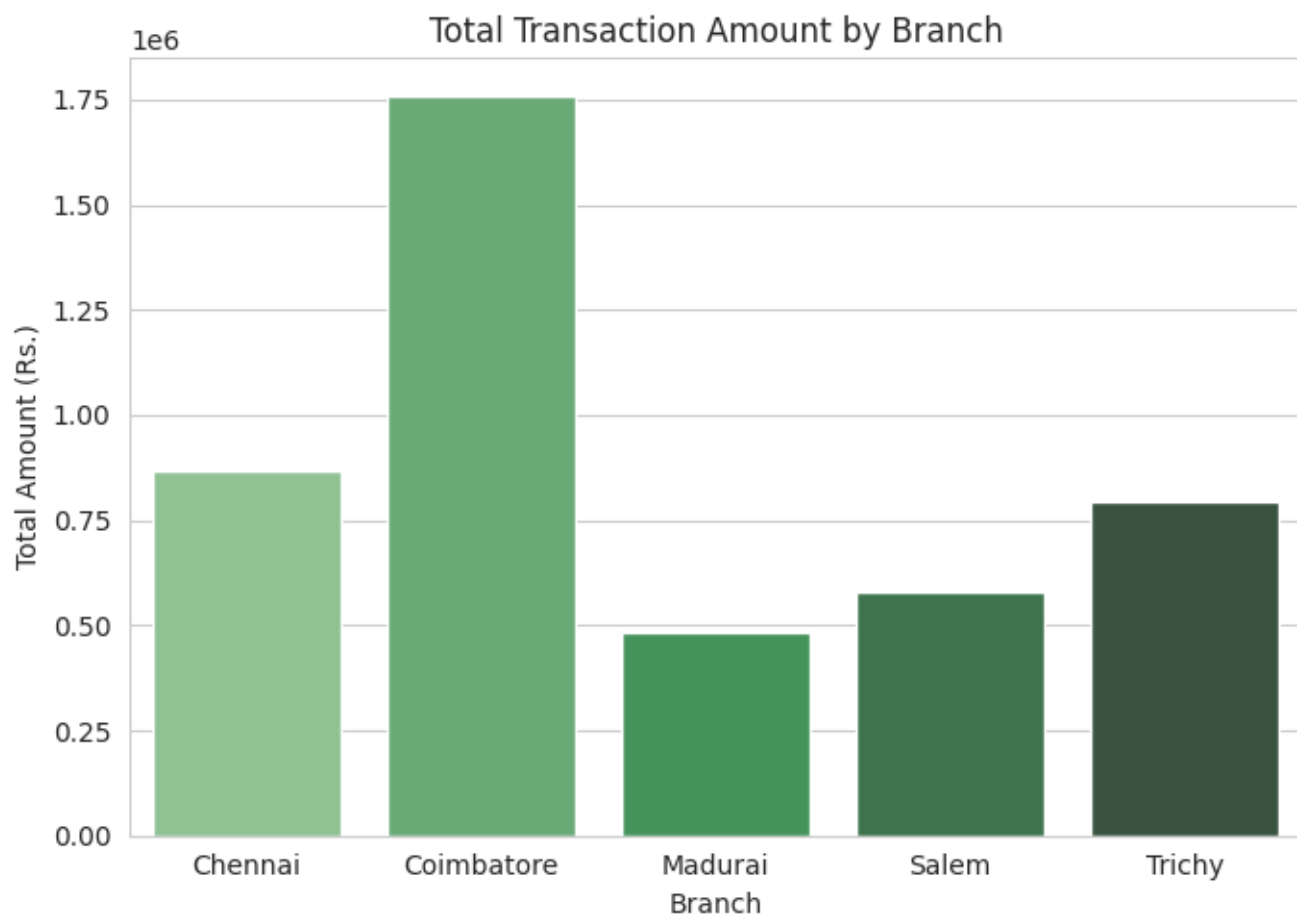
plt.figure(figsize=(9, 5))
monthly_trend.plot(kind="line", marker="o", color="#C44E52")
plt.title("Monthly Transaction Amount Trend (2024)")
plt.ylabel("Total Transaction Amount (Rs.)")
plt.xlabel("Month")
plt.tight_layout()
plt.show()

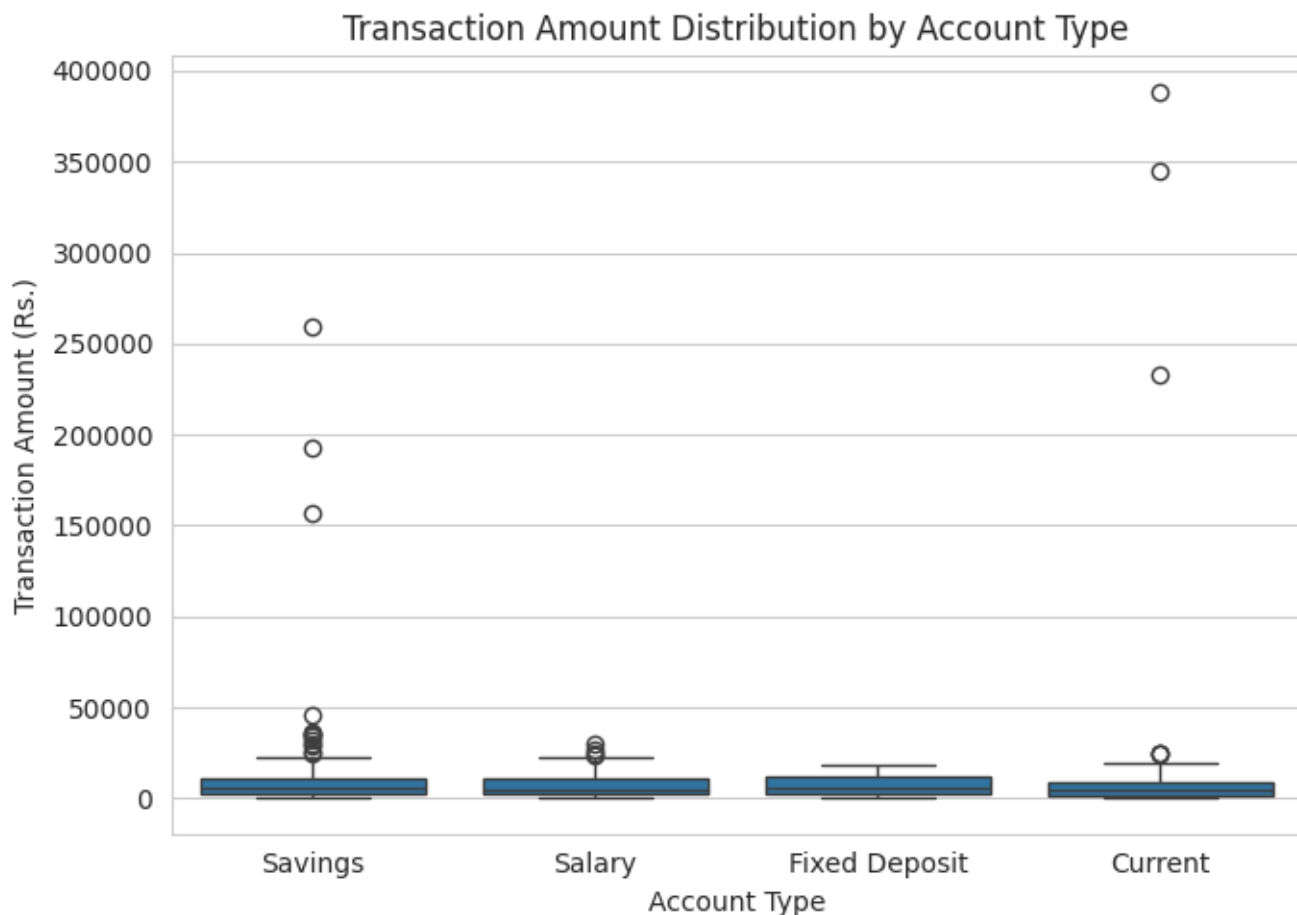
plt.figure(figsize=(7, 5))
sns.boxplot(x=df["Account_Type"], y=df["Transaction_Amount"])
plt.title("Transaction Amount Distribution by Account Type")
plt.ylabel("Transaction Amount (Rs.)")
plt.xlabel("Account Type")
plt.tight_layout()
plt.show()

```

Output:







Analysis

Transaction amounts ranged from about ₹9 to ₹3,88,609, with a median of about ₹5,458. Using the IQR method, any transaction above ₹22,281 was flagged as unusually large — 23 such transactions were found, several exceeding ₹1,50,000, concentrated in the Coimbatore and Trichy branches. Savings accounts recorded the highest total transaction value (₹20.1 lakh) followed by Current accounts (₹15.8 lakh), while Fixed Deposit accounts had the fewest transactions. Coimbatore branch handled the highest total transaction value overall. The correlation between how often a customer transacts and their average transaction amount was very weak ($r = -0.08$), suggesting frequent customers do not necessarily transact in larger amounts. The monthly trend showed peaks in January and February and a dip around April, July and September.

Solution Design

The bank should (1) route the 23 flagged large transactions through its fraud/AML review process, (2) focus relationship-management effort on Savings and Current account holders since they drive the highest transaction value, (3) investigate the Coimbatore branch's concentration of large transactions, and (4) plan staffing and liquidity around the January–February peak period identified in the monthly trend.

Clarity

Results are presented account-type-wise, branch-wise and month-wise with an explicit outlier threshold, making the findings directly usable for the bank's risk and customer-analytics teams.

Question 3: E-Commerce Product Analysis

Understanding of the Case

An e-commerce company wants to identify its best-performing products using data collected from separate sources: a product master (ProductID, Category, Price), sales records (ProductID, CustomerID, Quantity, OrderDate), and product reviews (ProductID, Rating, ReviewCount). These sources were integrated into a single table using Pandas merge operations. The resulting integrated dataset (ecommerce_product_sales.csv) contains 534 order-level rows across 60 products in six categories.

Application of Concepts

pandas.merge() was used to join the sales table with the product master on ProductID and then with the review table on ProductID, producing one combined table with price, rating and review count attached to every sale. A new column, SaleValue (Quantity × Price), was derived. groupby('Category') was used to compare categories, corr() was used to study relationships between Price, Rating, ReviewCount and SaleValue, and the IQR method was applied on total sale value per product to detect unusually high- or low-performing products.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
df = pd.read_csv("ecommerce_product_sales.csv")
cat_group = df.groupby("Category")["SaleValue"].sum().sort_values(ascending=False)
corr_matrix = df[["Price", "Rating", "ReviewCount", "SaleValue"]].corr()
product_sales = df.groupby("ProductID").agg(
    Total_Sale_Value=("SaleValue", "sum")
)
top5 = product_sales.sort_values("Total_Sale_Value", ascending=False).head(5)

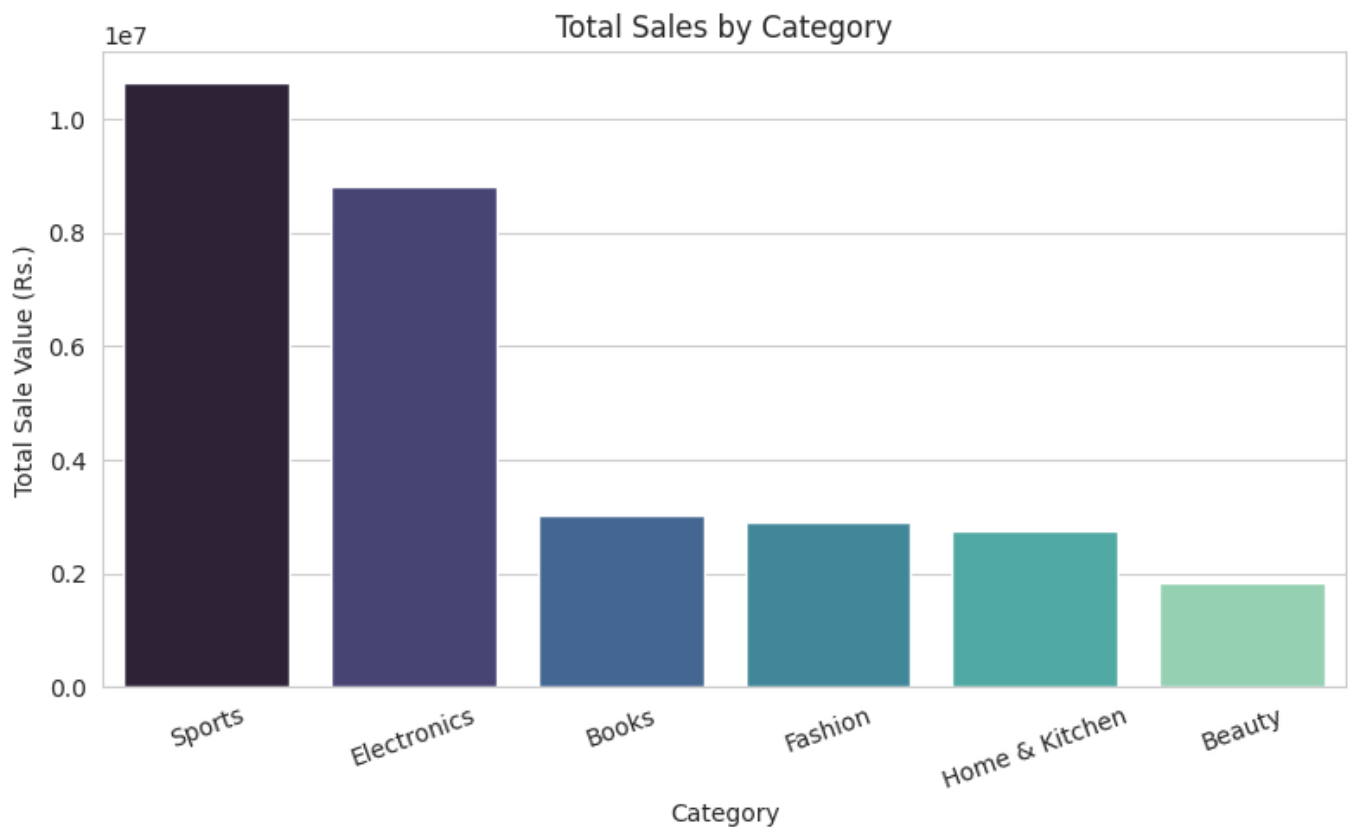
plt.figure(figsize=(8, 5))
sns.barplot(x=cat_group.index, y=cat_group.values, hue=cat_group.index, palette="mako", legend=False)
plt.title("Total Sales by Category")
plt.ylabel("Total Sale Value (Rs.)")
plt.xlabel("Category")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 6))
sns.scatterplot(data=df, x="Rating", y="SaleValue", hue="Category", alpha=0.7)
plt.title("Product Rating vs Sale Value")
plt.xlabel("Rating")
plt.ylabel("Sale Value (Rs.)")
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(8, 5))
sns.barplot(x=top5["Total_Sale_Value"], y=top5.index, hue=top5.index, palette="rocket", orient="h",
            legend=False)
plt.title("Top 5 Products by Total Sale Value")
plt.xlabel("Total Sale Value (Rs.)")
plt.ylabel("Product ID")
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(5.5, 4.5))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", vmin=-1, vmax=1, fmt=".2f")
plt.title("Correlation: Price, Rating, ReviewCount, SaleValue")
plt.tight_layout()
plt.show()
Output:
```



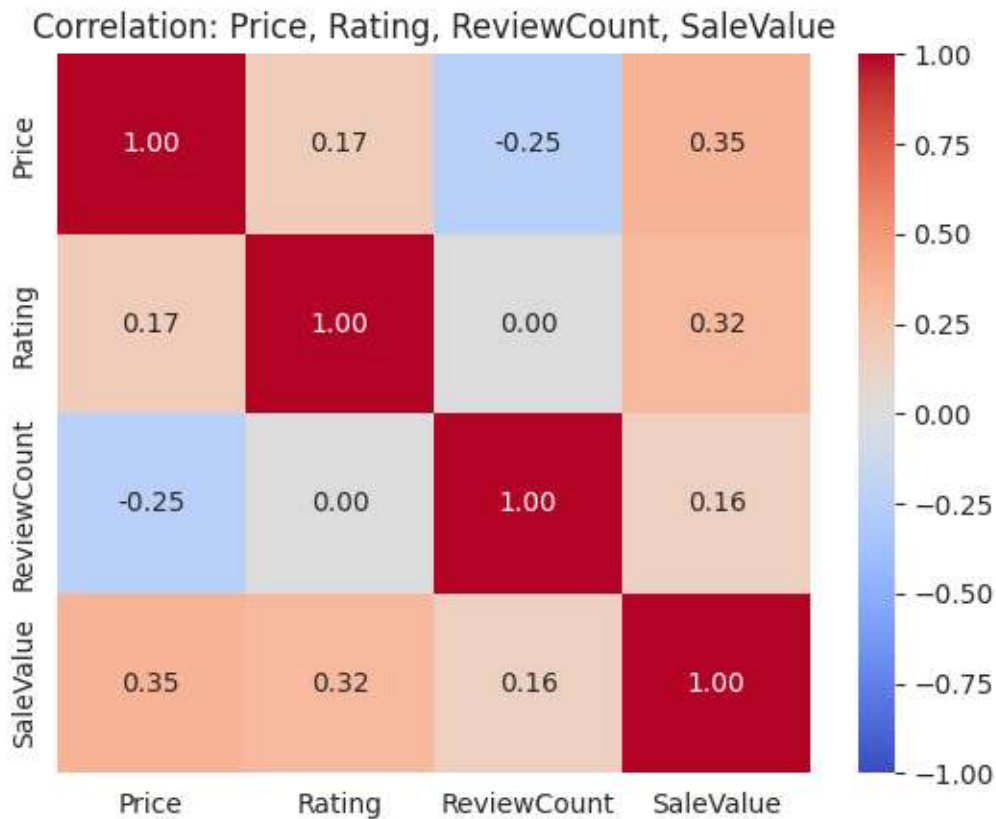
The scatter plot displays the distribution of product ratings across six categories. The X-axis represents the Rating (3.0 to 5.0), and the Y-axis represents the count of products (0 to 700,000). The legend indicates the following categories:

- Fashion (Blue)
- Electronics (Orange)
- Sports (Green)
- Beauty (Red)
- Home & Kitchen (Purple)
- Books (Brown)

The plot shows that most products have ratings between 3.5 and 4.5, with a significant peak at 5.0 for Electronics and Sports categories.

A horizontal bar chart titled 'Product Sales' showing the total sale value for five products. The y-axis lists the product IDs: P1002, P1044, P1005, P1009, and P1026. The x-axis represents the 'Total Sale Value (Rs.)' in millions, ranging from 0 to 8. The bars are color-coded in a gradient from dark purple for the highest value to light orange for the lowest values.

Product ID	Total Sale Value (Rs.) (in millions)
P1002	7.6
P1044	3.1
P1005	2.1
P1009	0.8
P1026	0.8



Analysis

No missing values were found after merging. Sports (₹1.07 crore) and Electronics (₹0.88 crore) generated the highest total sales among the six categories, while Beauty generated the least (₹18.4 lakh). The correlation analysis showed SaleValue is moderately related to Price ($r = 0.35$) and Rating ($r = 0.32$), and only weakly related to ReviewCount ($r = 0.16$); interestingly, Price and ReviewCount were slightly negatively correlated ($r = -0.25$), suggesting cheaper products attract more reviews. Using the IQR method on total sale value per product, three products (P1002, P1044, P1005) were identified as unusually high performers, together contributing a disproportionate share of total revenue, while no product fell below the low-performance threshold.

Solution Design

The company should (1) feature the three top-performing products (P1002, P1044, P1005) more prominently and ensure adequate stock, (2) study what makes these products successful (price point, rating) and apply similar pricing/marketing to other products in Sports and Electronics, (3) investigate low-rated, low-review products in Home & Kitchen and Books for potential quality or listing issues, and (4) use the mild negative correlation between price and review count to guide pricing strategy for products that need more customer feedback.

Clarity

The three separate data sources are clearly integrated before analysis, and the category-wise and product-wise summaries with an explicit outlier rule make the recommendations easy to follow.

Question 4: Weather Data Analysis

Understanding of the Case

A meteorological organization records daily/weekly weather readings — City, Date, Temperature, Humidity, Rainfall and Wind Speed — for several cities. The objective is to study temperature and rainfall patterns over time, compare cities and months, detect unusual readings, and examine correlations among the weather variables. The dataset used (weather_data.csv) contains 265 weekly readings for five cities across the year 2024.

Application of Concepts

The Date column was converted to datetime and a Month column was derived using dt.month_name(). groupby('City') and groupby('Month') were used to compare average temperature, humidity and total rainfall. The IQR method was applied separately to Temperature and Rainfall to flag unusual readings, and corr() was used to examine relationships between Temperature, Humidity, Rainfall and Wind Speed.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
df = pd.read_csv("weather_data.csv")
df["Date"] = pd.to_datetime(df["Date"])
df["Month"] = df["Date"].dt.month_name()
month_order = ["January", "February", "March", "April", "May", "June",
               "July", "August", "September", "October", "November", "December"]
monthly_city_temp = df.groupby(["Month", "City"])["Temperature_C"].mean().reset_index()
monthly_city_temp["Month"] = pd.Categorical(monthly_city_temp["Month"], categories=month_order,
ordered=True)
monthly_city_temp = monthly_city_temp.sort_values("Month")
month_rainfall = df.groupby("Month")["Rainfall_mm"].sum().reindex(month_order)
corr_matrix = df[["Temperature_C", "Humidity_%", "Rainfall_mm", "Wind_Speed_kmph"]].corr()
plt.figure(figsize=(10, 5))
sns.lineplot(data=monthly_city_temp, x="Month", y="Temperature_C", hue="City", marker="o")
plt.title("Monthly Average Temperature by City")
plt.ylabel("Temperature (C)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

```

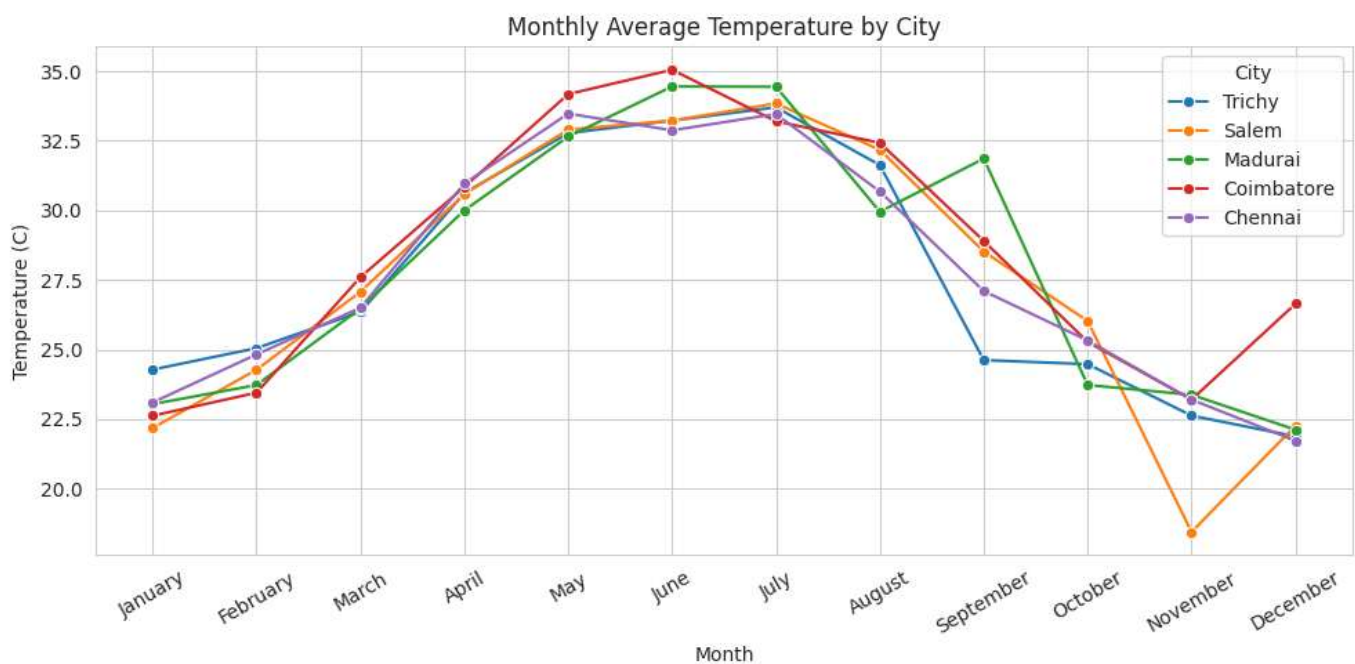
plt.figure(figsize=(9, 5))
sns.barplot(x=month_rainfall.index, y=month_rainfall.values, hue=month_rainfall.index,
palette="Blues_d", legend=False)
plt.title("Total Rainfall by Month")
plt.ylabel("Rainfall (mm)")
plt.xlabel("Month")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(x="City", y="Temperature_C", data=df)
plt.title("Temperature Distribution by City")
plt.ylabel("Temperature (C)")
plt.tight_layout()
plt.show()

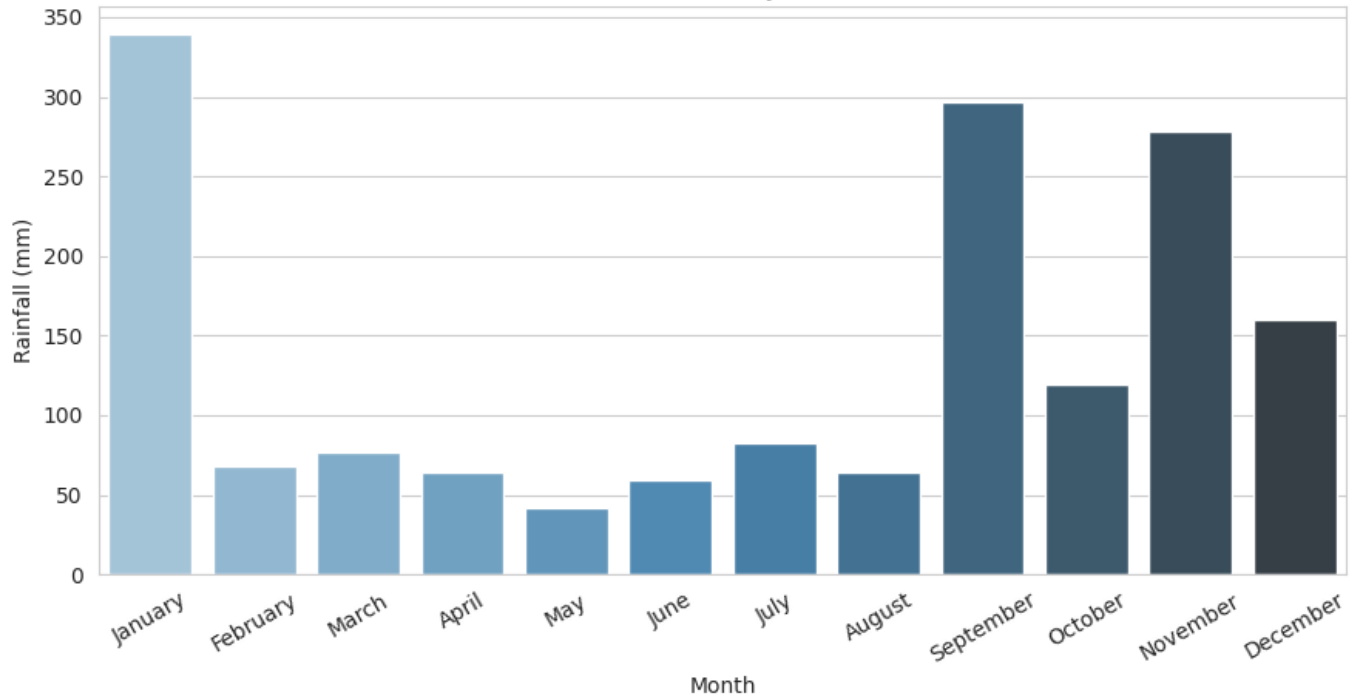
plt.figure(figsize=(5.5, 4.5))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", vmin=-1, vmax=1, fmt=".2f")
plt.title("Correlation: Temperature, Humidity, Rainfall, Wind Speed")
plt.tight_layout()
plt.show()

```

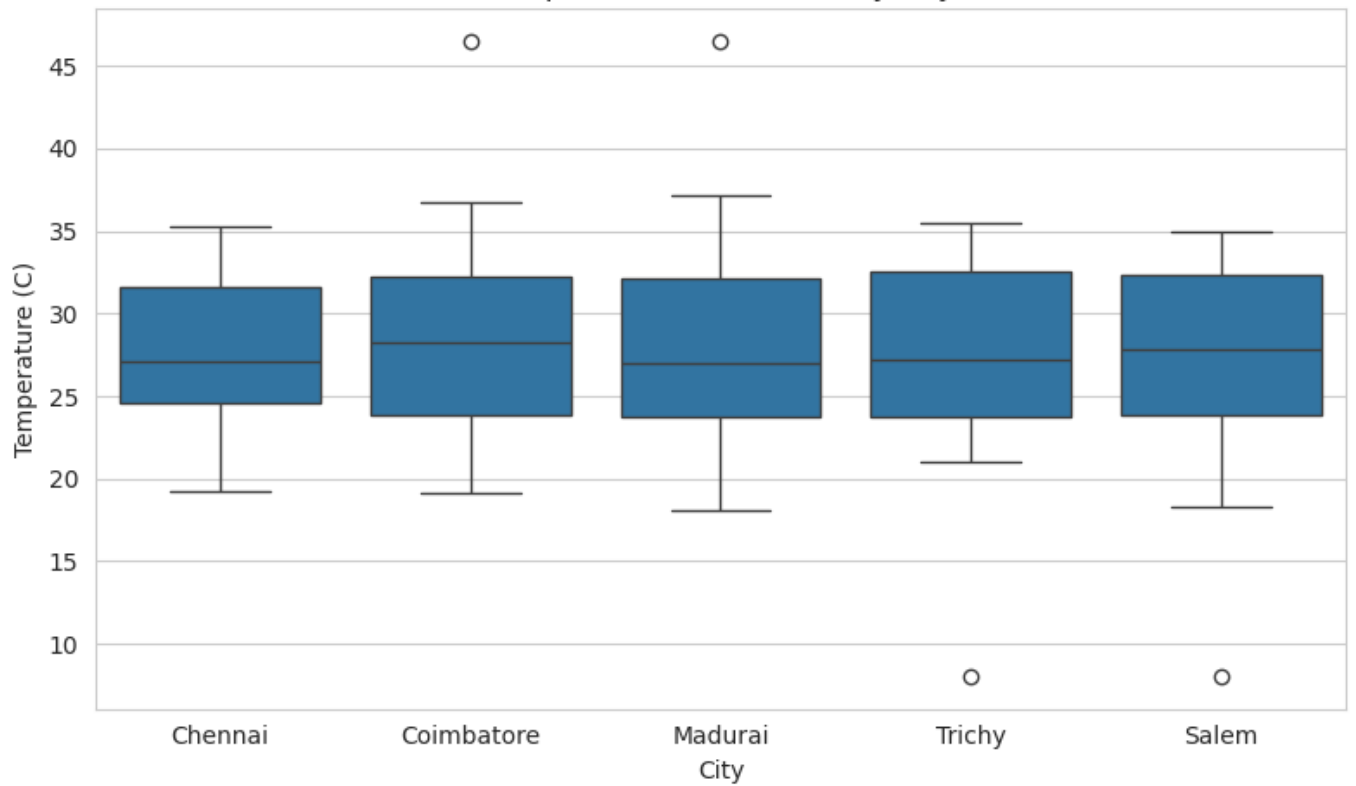
Output:

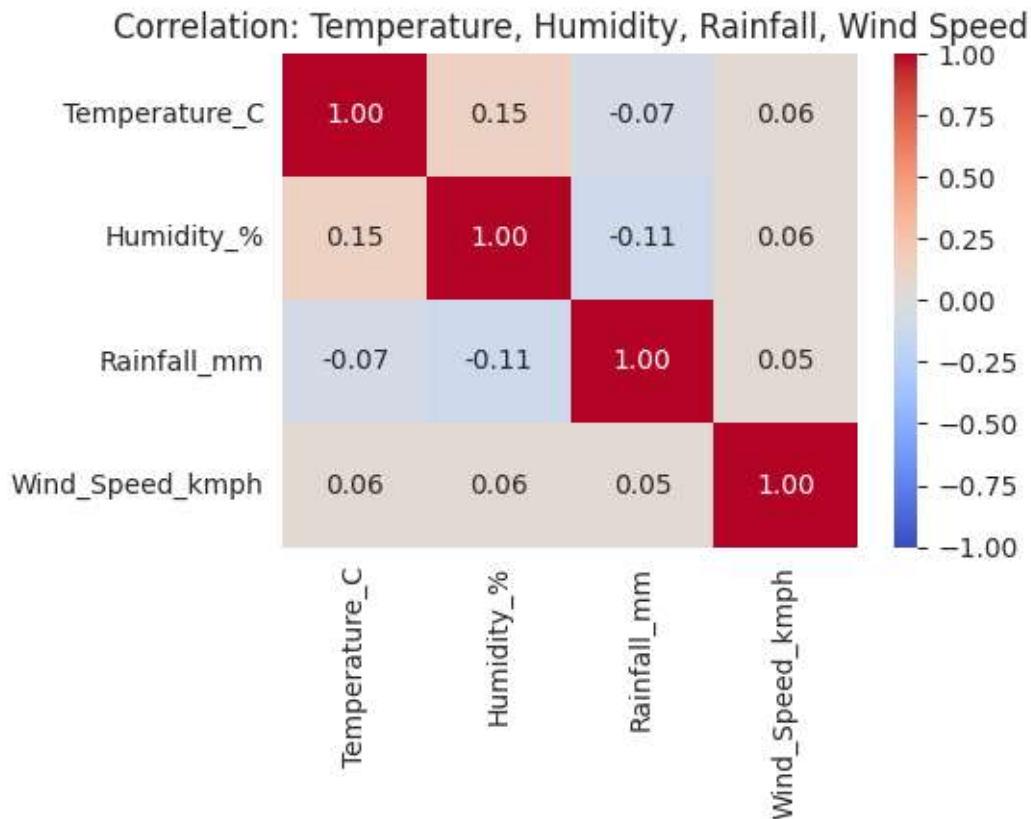


Total Rainfall by Month



Temperature Distribution by City





Analysis

Average temperature across cities was fairly uniform (27.5–28.6°C), while total rainfall varied considerably — Trichy recorded the highest total rainfall (450.7 mm) and Chennai the lowest (154.0 mm) over the year. Monthly grouping showed January, September and November as the wettest months and May–August as the hottest and driest. Four temperature readings were flagged as unusual by the IQR method — two unusually high (46.5°C in Coimbatore and Madurai) and two unusually low (8.0°C in Trichy and Salem) — along with several unusually heavy rainfall readings, the most extreme being 178.7 mm in Trichy and 163.0 mm in Salem. The correlation matrix showed only weak relationships among Temperature, Humidity, Rainfall and Wind Speed (all $|r| \leq 0.15$), indicating these variables move largely independently of one another in this dataset.

Solution Design

The organization should (1) verify the four flagged temperature readings and the extreme rainfall days against raw sensor logs to rule out instrument error, (2) issue advance flood-preparedness alerts for Trichy and Madurai/Salem around the wet months identified (January, September, November), (3) plan water conservation measures for Chennai given its comparatively low annual rainfall, and (4) continue monitoring humidity and wind speed independently of temperature since they show little linear relationship in this data.

Clarity

The city-wise and month-wise groupings, combined with a clearly stated outlier threshold and correlation summary, present the seasonal weather pattern in a structured and easy-to-follow manner.

https://colab.research.google.com/drive/1L1f2BniTBAKuzA6TiNx2KuLxXAu4QsJ_?usp=sharing